primepayments

MERCHANT API DOCUMENT

APPENDIX A & B

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Document Release Note

Version	Date	Description	Author
1.0.0	31 Aug 2020	First version	K5
1.0.1	27 Jan 2021	Add Java version of Message Encryption Example	K5

Appendix A - Message Encryption

Encryption Method : AES128
Block Cipher Mode : CBC

Merchant will be provided the followings when merchant account is ready

1) Merchant Key (16 characters)
2) Merchant Hash IV (16 characters)

1) Encrypt the message string (JSON Format) with AES128, cipher mode CBC
2) Use Base64 to encode the AES128 encrypted text

Example

PHP 7

```
$content = '{
      "merchantId": "94b2522ba609909d",
      "transactionId": "20201030023201612",
      "transactionType": "D",
      "payMethod": "offline",
      "currency": "RMB",
      "amount": "500.00",
      "callback": "https://enmu1r0iut2ws.x.pipedream.net",
      "response": "https://enmu1r0iut2ws.x.pipedream.net",
}';

$secret_key = $merchantKey
$secret_iv = $merchantHashIV;
$mode = "AES-128-CBC";
$encrypted = openssl_encrypt($content, $mode, $secret_key, OPENSSL_RAW_DATA, $secret_iv);
$message = base64_encode($encrypted);
```

JAVA 8

```
import javax.crypto.Cipher;
import javax.crypto.SecretKey;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import javax.crypto.IllegalBlockSizeException;
import javax.crypto.BadPaddingException;
import javax.crypto.NoSuchPaddingException;
import java.security.NoSuchAlgorithmException;
import java.security.InvalidKeyException;
import java.security.InvalidAlgorithmParameterException;
import java.io.UnsupportedEncodingException;
import java.util.Base64;

String content = "{" +
  "\"merchantId\": \"94b2522ba609909d\"," +
  "\"transactionId\": \"20201030023201612\"," +
  "\"transactionType\": \"D\"," +
  "\"payMethod\": \"offline\"," +
  "\"currency\": \"RMB\"," +
  "\"amount\": \"500.00\"," +
  "\"callback\": \"https://enmu1r0iut2ws.x.pipedream.net\"," +
  "\"response\": \"https://enmu1r0iut2ws.x.pipedream.net\"" +
"}";

String secretKey = "abcdefghhgfedcba";
String secretIv = "1234567887654321"; //Merchant Pass1 + Pass2

try {
  Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
  SecretKeySpec keySpec = new SecretKeySpec(secretKey.getBytes("utf-8"), "AES");
  IvParameterSpec ivSpec = new IvParameterSpec(secretIv.getBytes("utf-8"));
  cipher.init(Cipher.ENCRYPT_MODE, keySpec, ivSpec);
  byte[] ciphertext = cipher.doFinal(content.getBytes("UTF-8"));
  String message = Base64.getUrlEncoder().encodeToString(ciphertext);

  System.out.println("message=" + message);
} catch (NoSuchAlgorithmException e) {
  System.err.println(e.getMessage());
} catch (NoSuchPaddingException e) {
```

```
  System.err.println(e.getMessage());
} catch (InvalidAlgorithmParameterException e) {
  System.err.println(e.getMessage());
} catch (BadPaddingException e) {
  System.err.println(e.getMessage());
} catch (UnsupportedEncodingException e) {
  System.err.println(e.getMessage());
} catch (InvalidKeyException e) {
  System.err.println(e.getMessage());
} catch (IllegalBlockSizeException e) {
  System.err.println(e.getMessage());
}
```

Appendix B - Message Decryption

Decryption Method : AES128
Block Cipher Mode : CBC

Merchant will be provided the followings when merchant account is ready

1) Merchant Key (16 characters)
2) Merchant Hash IV (16 characters)

1) Use Base64 to decode the message to return raw message in bytes
2) Decrypt the raw message in bytes with AES128, cipher mode CBC to obtain the JSON plain data.

Example

PHP 7

```
$request = '{
      "merchantId":"82164414b30aa575",
      "message":"Eis\/kISbtVtkLckK\/LHMQQ1kZ9yy0oZV3tk0CPDJv5xfT+xPvbDT510cPL4b8QedGBsWkHqRmpDfbIGyWZVSB3egVdEEXW6I8q6v
xhccIoHruCclCzrIvP9e\/Y1ick1gkMAEZRCTs5jLLvD3waDU+nzMGIyY2oGtn0V0Lu+bSVAjigRDBrwxnullPU2ABvzh9jq4ARAwxjJQPBVIAAeaWTNsqR
i1dvMo4xp6Oq1i9rUafg4DREkizh\/h4p+1a+YwvluuiMnZC4RcCHsiW5CyjPd+Tg6bSkEYFm4Ddh8bNgLgxxlM9U7BTwX2mvSRnLoZotTgflVAva20LiGc
PeM3AllcPuma5khdlAmxaTnFc3uaoKtWPAPEElgldcEZvh\/2Mlra+2t2MfegpcDpquFwr0+Fm+9ZUg2zQUjAx7ry3N6DuFMlKMTJqxs3\/d\/EXvqI\/gL
8K9ntXMnC229140u71UvZyv+jfDr7mgsSwurW5rsIDpZadwKezEsv7Y7KBe9xGPuh+jNev+SqJKytvoG6xwttnqy+K6IPiLFLwXs54GlhKizDVnIamp2cJp
KeGeH+V7C84jzX5f+dXf+yTDkAffUjpD4b0heJM\/7PMwdZqbFho7KHKZyR9OyHkXcHa2J2jN8FB\/egBh0PWncQutxswGUk1+maQWdZkZM5KzF0SRc50kC
vGZPi2vgvKZMcybnXfOttO2Z+PTmd5b0nOtuo5w=="
}';

$content = json_encode($request);
$content = $content->message;

$secret_key = $merchantKey
$secret_iv = $merchantHashIV;
$mode = "AES-128-CBC";
$encrypted = base64_decode($content);
$message = openssl_decrypt($encrypted, $mode, $secret_key, OPENSSL_RAW_DATA, $secret_iv);
```

JAVA 8

```
import javax.crypto.Cipher;
import javax.crypto.SecretKey;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import javax.crypto.IllegalBlockSizeException;
import javax.crypto.BadPaddingException;
import javax.crypto.NoSuchPaddingException;
import java.security.NoSuchAlgorithmException;
import java.security.InvalidKeyException;
import java.security.InvalidAlgorithmParameterException;
import java.io.UnsupportedEncodingException;
import java.util.Base64;

String requestMessage =
  "dO_ih-LJC7fpBjKRTr0Hbmw0rPvn4MKP3XwDPWxPYFY7T09dEze-tar7NXdYqJ0kY-osQs822lyks-
SI3s9NCidEtbulRDtDeBF2Tq0lhMAGseEDgJnyHu6CtSTLCM3iecKYX9qfsK64GndX1wrIAL2mMPeiWNfYN14rowLliqQU8TkvVR2Rd6Ml2XpecL5YXAUsG
Tq8omxFb3YBB5aYoU9n1rh-NzL_43IQJQw-Tx4Kc-eLniTwNJgVUbpYox3vFjrbVeF3pNFZ832nWnXB7RZ0FLCdz77CUXJ4sQdS17z7JgwXmCPxD22-
YybQixlAbP-tSPIebs1i44qiSo7b9asG7d-y20A7ePk3qgLj_PQ=";

String content = "{" +
  "\"merchantId\": \"94b2522ba609909d\"," +
  "\"message\": \"" + requestMessage + "\"" +
"}";

String secretKey = "abcdefghhgfedcba";
String secretIv = "1234567887654321"; //Merchant Pass1 + Pass2

try {
  Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
  SecretKeySpec keySpec = new SecretKeySpec(secretKey.getBytes("utf-8"), "AES");
  IvParameterSpec ivSpec = new IvParameterSpec(secretIv.getBytes("utf-8"));
  cipher.init(Cipher.DECRYPT_MODE, keySpec, ivSpec);
  byte[] requestMessageBytes = Base64.getUrlDecoder().decode(requestMessage);
  byte[] plainText = cipher.doFinal(requestMessageBytes);
  String message = new String(plainText );
```

```
  System.out.println("message=" + message);
} catch (NoSuchAlgorithmException e) {
  System.err.println(e.getMessage());
} catch (NoSuchPaddingException e) {
  System.err.println(e.getMessage());
} catch (InvalidAlgorithmParameterException e) {
  System.err.println(e.getMessage());
} catch (BadPaddingException e) {
  System.err.println(e.getMessage());
} catch (UnsupportedEncodingException e) {
  System.err.println(e.getMessage());
} catch (InvalidKeyException e) {
  System.err.println(e.getMessage());
} catch (IllegalBlockSizeException e) {
  System.err.println(e.getMessage());
}
```